

Homoscedasticity and Its Testing: From Conditional Expectation to the Breusch–Pagan Test

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Part 1 / 12
Conditional Expectation and the Regression Function

July 2026

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1 Homoscedasticity in Elementary Regression Textbooks

Three settings are usually conflated in introductory regression courses: Y_i 's with fixed indices, the linear model with non-random X , and the joint-distribution setup with random X . All three talk about “homoscedasticity”, but they mean different things, and the slippage between them is a frequent source of confusion. This chapter separates them cleanly before moving on to the measure-theoretic formulation used in the rest of the text.

1.1 Non-random X : homoscedasticity as equal variances

In introductory regression courses, the regressor X is treated as non-random: it is a fixed set of index values $x_1, \dots, x_n \in \mathbb{R}$. For each index x_i there is a target random variable Y_i , all defined on the same probability space. We can write this as

$$Y_{x_1}, Y_{x_2}, \dots, Y_{x_n},$$

and if we drop the strange subscripts, we simply have a random vector $(Y_1, \dots, Y_n)$ — n random variables on a common space.

Definition 1. In the non-random- X setting, the variables $Y_1, \dots, Y_n$ are called **homoscedastic** if

$$\text{Var}(Y_1) = \text{Var}(Y_2) = \dots = \text{Var}(Y_n).$$

This is a definition by fiat: equal marginal variances. It does not yet involve a model or errors.

Remark 2. This can be taken as the definition, but the Probabilist's view is that it is better understood as a *consequence* of a more primitive assumption about the errors, as shown in Section 1.2.

1.2 The linear model with non-random X

The standard linear model is

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n,$$

where $x_1, \dots, x_n$ are fixed numbers and $\varepsilon_1, \dots, \varepsilon_n$ are random errors. Here the x_i are not random—they are parameters of the model, not outcomes of a random variable.

Homoscedasticity in this model means

$$\text{Var}(\varepsilon_1) = \text{Var}(\varepsilon_2) = \dots = \text{Var}(\varepsilon_n) = \sigma^2.$$

This implies $\text{Var}(Y_i) = \sigma^2$ for all i , since $\beta_0 + \beta_1 x_i$ is a constant.

Independence structure. If $\varepsilon_1, \dots, \varepsilon_n$ are independent, then $Y_1, \dots, Y_n$ are also independent. However, i.i.d. errors do *not* imply i.i.d. responses:

$$\mathbb{E}Y_i = \beta_0 + \beta_1 x_i,$$

so the Y_i have different means whenever the x_i differ. Independent with unequal distributions is not i.i.d.

Remark 3. This contrast matters: in the non-random- X setting, “identically distributed” is simply not the right concept for the Y_i . The meaningful homogeneity condition is constancy of *variance*, not of distribution.

1.3 Random X : the joint distribution model

Now suppose (X, Y) is a random vector defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We observe an i.i.d. sample of pairs:

$$(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n),$$

where each pair (X_i, Y_i) has the same joint distribution as (X, Y) , and the pairs are mutually independent (in the full sense: the generated σ -algebras $\sigma(X_i, Y_i)$ are independent).

The i.i.d. paradox. In this model, $(X_1, Y_1) \stackrel{d}{=} (X_2, Y_2) \stackrel{d}{=} \dots$, and in particular $Y_1 \stackrel{d}{=} Y_2$. Yet in the non-random- X model, Y_1 and Y_2 have different distributions (different means). What is going on?

The resolution is that these are genuinely different models. In the random- X model, each (X_i, Y_i) is an independent draw from the *same* joint distribution; the X_i vary randomly, and the marginal distribution of Y_i is the same for all i by symmetry. In the non-random- X model, the x_i are fixed constants; conditioning on specific values $x_1 \neq x_2$ selects slices of the joint distribution with different conditional means.

Example 4. Consider a single random variable X with sample (X_1, X_2) i.i.d. $\sim X$. Before observing, $X_1 \stackrel{d}{=} X_2$. Once we observe the outcomes (x_1, x_2) , we might think of X_1 and X_2 as degenerate: $X_1 \sim \delta_{x_1}$, $X_2 \sim \delta_{x_2}$. If $x_1 \neq x_2$ these are different distributions—no longer i.i.d. The non-random- X regression model is precisely this: we condition on the observed x_i , and the i.i.d. structure of the pairs collapses.

This is not a paradox but a model choice. Fixing X (conditioning) and treating X as random are different setups, with different implications for what “i.i.d.” means.

2 Conditional Expectation: Definition and Properties

2.1 Definition

Let (X, Y) be a random vector on $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{E}|Y| < \infty$. A random variable $Z = Z(\omega)$ is called $\mathbb{E}[Y \mid X]$, the **conditional expectation of Y given X** , if:

- (1) Z is $\sigma(X)$ -measurable, and
- (2) $\int_A Z \, d\mathbb{P} = \int_A Y \, d\mathbb{P}$ for all $A \in \sigma(X)$.

Remark 5 (Operator view). The conditional expectation is a binary operator taking two random variables X and Y as arguments and returning a new random variable $Z = \mathbb{E}[Y \mid X]$. More precisely:

$$\mathbb{E}[\cdot \mid \cdot] : (X, Y) \longmapsto Z,$$

where the output Z is itself a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$, with values in $\mathbb{R}$. The same operator, applied to the same pair (X, Y) but with a squared-residual argument, gives the conditional variance (the Conditional Variance section (Part 6)).

When we pass to the Doob–Dynkin representation $Z = h(X)$, we are factoring this operator through the values of X : the function $h : \mathbb{R} \rightarrow \mathbb{R}$ encodes the same information as Z , but as a function of the *realised value* of X rather than of ω directly.

2.2 Existence, uniqueness, and distribution

Proposition 6 (Existence). *Under $\mathbb{E}|Y| < \infty$, a random variable Z satisfying (1)–(2) always exists.*

Sketch. This is a consequence of the Radon–Nikodym theorem. Define a signed measure $\nu(A) = \int_A Y d\mathbb{P}$ on $\sigma(X)$. Since Y is integrable, ν is absolutely continuous with respect to $\mathbb{P}|_{\sigma(X)}$. By Radon–Nikodym, there exists a $\sigma(X)$ -measurable density Z such that $\nu(A) = \int_A Z d\mathbb{P}$ for all $A \in \sigma(X)$. This Z satisfies (1) and (2). Reference: Rudin, *Real and Complex Analysis*, Theorem 6.10; Kallenberg, *Foundations of Modern Probability*, Theorem 5.1. $\square$

Proposition 7 (Uniqueness up to null sets). *If Z_1 and Z_2 both satisfy (1)–(2), then $Z_1 = Z_2$ almost surely.*

Proof. Both satisfy condition (2), so $\int_A Z_1 d\mathbb{P} = \int_A Z_2 d\mathbb{P}$ for all $A \in \sigma(X)$, hence $\int_A (Z_1 - Z_2) d\mathbb{P} = 0$. Set $A = \{Z_1 > Z_2\} \in \sigma(X)$. On A the integrand $Z_1 - Z_2 > 0$, so $\mathbb{P}(A) = 0$. Similarly $\mathbb{P}(Z_1 < Z_2) = 0$. Hence $\mathbb{P}(Z_1 \neq Z_2) = 0$. $\square$

Remark 8 (Distribution). Since any two versions of $\mathbb{E}[Y | X]$ agree almost surely, they induce the same pushforward measure on $\mathbb{R}$. That is, all versions have the same distribution. This follows immediately from uniqueness: if $Z_1 = Z_2$ a.s., then $\mathbb{P}(Z_1 \in B) = \mathbb{P}(Z_2 \in B)$ for every Borel set B .

2.3 The equivalence theorem

The following theorem gives an alternative characterisation of the conditional expectation that follows directly from the definition, without any use of the Doob–Dynkin factorisation.

Proposition 9 (Equivalence theorem). *Let Z be $\sigma(X)$ -measurable with $\mathbb{E}|Z| < \infty$, and set $u = Y - Z$. Then:*

$$Z = \mathbb{E}[Y | X] \iff \mathbb{E}[u | X] = 0.$$

Proof. $(\Rightarrow)$ Assume $Z = \mathbb{E}[Y | X]$. Then:

$$\mathbb{E}[u | X] = \mathbb{E}[Y - Z | X] = \mathbb{E}[Y | X] - \mathbb{E}[Z | X] = Z - Z = 0,$$

where the last equality uses that Z is $\sigma(X)$ -measurable, so $\mathbb{E}[Z | X] = Z$.

$(\Leftarrow)$ Assume $\mathbb{E}[u | X] = 0$. Rearrange: $Z = Y - u$. Taking conditional expectations:

$$\mathbb{E}[Z | X] = \mathbb{E}[Y - u | X] = \mathbb{E}[Y | X] - \mathbb{E}[u | X] = \mathbb{E}[Y | X].$$

Since Z is $\sigma(X)$ -measurable, $\mathbb{E}[Z | X] = Z$, giving $Z = \mathbb{E}[Y | X]$. $\square$

Decomposition. Taking $Z = \mathbb{E}[Y | X]$ and $u = Y - Z$, we have $Y = Z + u$ with $\mathbb{E}[u | X] = 0$. This holds with no further assumptions beyond $\mathbb{E}|Y| < \infty$.

2.4 Doob–Dynkin factorisation and the regression function

The equivalence theorem above characterises $\mathbb{E}[Y | X]$ purely through the definition. The Doob–Dynkin lemma adds further structure: since $Z = \mathbb{E}[Y | X]$ is $\sigma(X)$ -measurable, there exists a Borel-measurable function $h : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$Z(\omega) = h(X(\omega)) \quad \text{for all } \omega \in \Omega.$$

The function h is the **regression function** of Y on X : $h(x) = \mathbb{E}[Y | X = x]$ for (almost) every x . The decomposition now reads $Y = h(X) + u$.

Degrees of freedom: Z vs. h .

Freedom of Z . By the uniqueness result (Section 4.2), any two versions Z_1, Z_2 of $\mathbb{E}[Y \mid X]$ agree $\mathbb{P}$ -almost surely. The freedom of Z is exactly: modification on $\mathbb{P}$ -null subsets of Ω .

Freedom of h . Any two Borel functions h_1, h_2 satisfying $h_1(X) = Z$ a.s. and $h_2(X) = Z$ a.s. must satisfy $h_1(X) = h_2(X)$ $\mathbb{P}$ -a.s., which means

$$\mathbb{P}(h_1(X) \neq h_2(X)) = 0, \quad \text{i.e.} \quad P_X(\{x : h_1(x) \neq h_2(x)\}) = 0.$$

So h is unique P_X -almost everywhere on $\mathbb{R}$: two versions may differ only on a P_X -null Borel set. The freedom of h is: modification on P_X -null subsets of $\mathbb{R}$.

Comparing the two freedoms. The relationship between the two measures is $P_X(B) = \mathbb{P}(X \in B)$ for every Borel $B \subseteq \mathbb{R}$. Therefore:

- A $\mathbb{P}$ -null set $N \subseteq \Omega$ corresponds, via X , to modifying h on $X(N) \subseteq \mathbb{R}$, which is a P_X -null set. So every freedom of Z translates into a freedom of h .
- Conversely, any Borel set $B \subseteq \mathbb{R}$ with $P_X(B) = 0$ gives a freedom of h (modify h on B arbitrarily), and the corresponding set $\{X \in B\}$ has $\mathbb{P}$ -measure zero, so this change is invisible to Z as well.
- However, h also has freedom on Borel sets B with $P_X(B) = 0$ that are *not* in the image of any ω under X —for instance, any open interval on which P_X assigns zero mass. For such B , modifying h on B cannot even be described in terms of Z : the event $\{X \in B\}$ has $\mathbb{P}$ -measure zero, so the modification has no counterpart on Ω at all.

This last point is the precise sense in which h has *strictly more* freedom than Z . The extra freedom consists of modifications on those P_X -null Borel sets in $\mathbb{R}$ for which $\mathbb{P}(X \in B) = 0$, including sets that X may reach on a $\mathbb{P}$ -null set of ω 's.

Remark 10 (Support and its complications). One is tempted to describe the extra freedom of h as “freedom outside the support of P_X .” The support of P_X on $\mathbb{R}$ is standardly defined as the smallest closed set of full P_X -measure, or equivalently $\mathbb{R} \setminus \bigcup \{U \text{ open} : P_X(U) = 0\}$. This is well-defined on $\mathbb{R}$ with the usual topology, because an arbitrary union of open P_X -null sets is again open and has P_X -measure zero.

However, using the support to delineate freedom is imprecise in two ways. First, the support can contain P_X -null subsets (individual points under a continuous distribution), and h is free on those too. Second, range and support are different: $X(\Omega)$ and $\text{supp}(P_X)$ need not coincide— X could take values outside the support on a $\mathbb{P}$ -null set of ω 's, or conversely the support could contain points that X never reaches. The cleanest statement is: h is determined P_X -almost everywhere on $\mathbb{R}$, and any Borel set of P_X -measure zero is a set of freedom for h .

The linear regression assumption. The standard linear model imposes $h(x) = \beta_0 + \beta_1 x$ on all of $\mathbb{R}$. This fixes h on P_X -null sets where it is otherwise undetermined—an extrapolation assumption that the data cannot test.

3 Measure-Dependent Quantities

This chapter collects the conceptual foundations that underlie the likelihood-based tests discussed later: the notion of a measure-dependent function, the distinction between a statistic and a random quantity, the push-forward construction, and a concrete illustration of iterating measure-dependent maps.

This section develops the conceptual framework needed to make the prediction story from the Consequences section (Part 6) precise. It clarifies the distinction between a statistic, a random quantity, and a measure-dependent function — a distinction that is usually left implicit in textbooks.

3.1 The problem with $\mathbb{E}[Y_2 \mid Y_1]$ as a predictor

From the i.i.d. forecast notes: given a single observation Y_1 , the MSE-optimal predictor of $Y_2 \stackrel{d}{=} Y_1$ is $f^*(Y_1) = \mathbb{E}[Y_2 \mid Y_1]$. But wait — can we actually use this as a predictor?

The conditional expectation $\mathbb{E}[Y_2 \mid Y_1]$ depends on the joint distribution P of (Y_1, Y_2) . It is not the same function for all measures. A predictor that requires knowing P to evaluate is not useful in practice, and is not a statistic in any standard sense.

3.2 Random quantities and statistics

Definition 11. Let $\{P_\theta : \theta \in \Theta\}$ be a parametric family of distributions on $\mathbb{R}^n$. A **random quantity** is a measurable function

$$f : \mathbb{R}^n \times \Theta \rightarrow \mathbb{R}, \quad (x_1, \dots, x_n, \theta) \mapsto f(x_1, \dots, x_n, \theta).$$

A random quantity f is called a **statistic** (with respect to this family) if it does not depend on θ :

$$f(x_1, \dots, x_n, \theta) = f(x_1, \dots, x_n, \theta') \quad \text{for all } \theta, \theta' \in \Theta.$$

Remark 12 (Two views on statistics). The Probabilist prefers the textbook convention: a statistic is simply a measurable function of the data, with no θ argument at all. The definition above is equivalent when the family is large enough — if the family contains all possible distributions, then any function that happens to not depend on θ trivially satisfies the definition. The explicit θ argument is useful when we want to *distinguish* statistics from non-statistics within a specific model family.

Remark 13 (Monotonicity with respect to the family). The statistic property is monotone with respect to inclusion of measure families.

- **Downward monotonicity.** If f is a statistic with respect to a family $\mathcal{P}$, then it is also a statistic with respect to every subfamily $\mathcal{Q} \subseteq \mathcal{P}$. A smaller family imposes fewer constraints on measure-independence, so any function that does not depend on the measure within $\mathcal{P}$ certainly does not depend on it within $\mathcal{Q}$.
- **Upward monotonicity.** If f is *not* a statistic with respect to $\mathcal{P}$ (i.e. it takes different values for some $P, P' \in \mathcal{P}$), then it is also not a statistic with respect to any larger family $\mathcal{Q} \supseteq \mathcal{P}$, since $\mathcal{Q}$ still contains the distinguishing pair P, P' .

In particular: if f is a statistic with respect to the full family of all probability measures, it is a statistic with respect to every subfamily (downward monotonicity applied to the largest possible family). Conversely, if f is *not* a statistic with respect to some family $\mathcal{P}$, it fails to be a statistic with respect to every superfamily $\mathcal{Q} \supseteq \mathcal{P}$. Being a statistic with respect to a singleton family $\{P\}$ is vacuous and tells us nothing about larger families.

Examples.

- $f(Y_1) = Y_1$: a statistic. It does not depend on P .
- $f_P(Y_1) = \mathbb{E}_P[Y_2 \mid Y_1]$: not a statistic. It depends on the joint distribution P of (Y_1, Y_2) and is different for different P .
- $h_P(X) = \mathbb{E}_P[Y \mid X]$ (regression function): not a statistic. It depends on the joint distribution of (X, Y) . To be precise, we should write h_P or $h_{P_{XY}}$.
- $\hat{\sigma}^2 = \frac{1}{n-2} \sum \hat{u}_i^2$: a statistic. Computable from data alone.

3.3 Mixed random quantities and approximation

Some objects depend on both the sample and the parameter — they are neither pure statistics nor pure parameters. Examples:

- $\mathbb{E}_P[Y_2 \mid Y_1]$: depends on Y_1 (data) and on P (measure).
- $\text{Var}(\hat{\beta}_1 \mid \mathbf{X}) = \sigma^2 / S_{XX}$: depends on $\mathbf{X}$ (data) and on σ^2 (parameter).
- $h_P(X_i) = \mathbb{E}_P[Y_i \mid X_i]$: depends on X_i (data) and on P (measure).

How do we approximate mixed quantities, and by what? In principle, any random quantity — statistic or not — can serve as an approximation. But in practice, only statistics are usable: a non-statistic requires knowing the measure to evaluate, which defeats the purpose. The plug-in principle is the standard way to convert a non-statistic into a statistic: replace the unknown measure-dependent object by an estimate.

On the best estimator of $\mathbb{E}_P Y$. Suppose we want to estimate the parameter $\theta_P = \mathbb{E}_P Y$ by a function g of data $X_1, \dots, X_n$. If we require g to be a true statistic (measure-independent), then the MSE depends on P and we must specify for which P we optimise.

If instead we allow g to depend on P , the optimum is trivial: set $g_P \equiv \theta_P$ (a constant equal to the parameter). This is formally a valid “estimator” — a constant is a function of the sample — but it is measure-dependent and thus not a statistic. It uses the parameter to estimate itself, which is circular.

This shows why requiring measure-independence (the statistic property) is not merely a convention: it prevents the degenerate solution of “estimating θ_P by θ_P .”

Remark 14 (Singleton family). If the family of measures is a singleton $\{P\}$, then every random quantity is trivially a statistic. In this case, any random quantity is a statistic. Is that right? Yes — and this also means the constant θ_P is a valid statistic in that case. The degenerate solution is only excluded when the family is non-trivial.

Remark 15 (Degenerate case). If the family of measures is a singleton $\{P\}$, then every random quantity is trivially a statistic: there is no θ to depend on. This is consistent with the definition but loses the distinction between parameters, statistics, and measure-dependent quantities.

3.4 MSE decomposition: the general case

Let P be a joint distribution of (Y_1, Y_2) with $\mathbb{E}_P[Y_2^2] < \infty$. For any random quantity f (possibly measure-dependent), the prediction MSE decomposes as follows.

Add and subtract the measure-specific optimal predictor $f_P^*(Y_1) = \mathbb{E}_P[Y_2 \mid Y_1]$:

$$\begin{aligned} \mathbb{E}_P[(f(Y_1) - Y_2)^2] &= \mathbb{E}_P[(f(Y_1) - f_P^*(Y_1)) + (f_P^*(Y_1) - Y_2)]^2 \\ &= \mathbb{E}_P[(f(Y_1) - f_P^*(Y_1))^2] + 2\mathbb{E}_P[(f(Y_1) - f_P^*(Y_1))(f_P^*(Y_1) - Y_2)] \\ &\quad + \mathbb{E}_P[(f_P^*(Y_1) - Y_2)^2]. \end{aligned}$$

The cross-term vanishes.

$$\begin{aligned} &\mathbb{E}_P[(f(Y_1) - f_P^*(Y_1)) \cdot (f_P^*(Y_1) - Y_2)] \\ &= \mathbb{E}_P[\mathbb{E}_P[(f(Y_1) - f_P^*(Y_1)) \cdot (f_P^*(Y_1) - Y_2) \mid Y_1]] \\ &= \mathbb{E}_P[(f(Y_1) - f_P^*(Y_1)) \cdot \mathbb{E}_P[f_P^*(Y_1) - Y_2 \mid Y_1]] \\ &= \mathbb{E}_P[(f(Y_1) - f_P^*(Y_1)) \cdot 0] = 0, \end{aligned}$$

where we used $\mathbb{E}_P[f_P^*(Y_1) - Y_2 \mid Y_1] = f_P^*(Y_1) - \mathbb{E}_P[Y_2 \mid Y_1] = 0$.

Conclusion.

$$\mathbb{E}_P[(f(Y_1) - Y_2)^2] = \underbrace{\mathbb{E}_P[(f(Y_1) - f_P^*(Y_1))^2]}_{\geq 0} + \underbrace{\mathbb{E}_P[(f_P^*(Y_1) - Y_2)^2]}_{\text{irreducible floor}}. \quad (1)$$

Denoting the measure-specific minimizer $f_P^*(Y_1) = \mathbb{E}_P[Y_2 \mid Y_1]$:

$$\min_f \mathbb{E}_P[(f(Y_1) - Y_2)^2] = \mathbb{E}_P[(f_P^*(Y_1) - Y_2)^2],$$

and f_P^* is a minimizer. But this minimizer depends on P — different measures give different optimal predictors.

3.5 The maxi-min problem

Since the optimal predictor f_P^* changes with P , a natural question is: can we find a single statistic f (measure-independent) that performs well uniformly across all measures?

One way to formalise this is the **maxi-min criterion**: find the measure P that is hardest to predict, and evaluate the best measure-specific predictor there:

$$\max_{P_{Y_1, Y_2}} \min_{f_P} \mathbb{E}_P[(f_P(Y_1) - Y_2)^2].$$

This is the worst-case irreducible floor over all joint distributions. It does not directly give a statistic, because both the max and the min are taken over measure-dependent objects.

Remark 16. This criterion does not appear in standard textbooks on prediction or regression. The usual approach is to fix a family of measures (e.g. i.i.d., or parametric time-series) and work within that family. The maxi-min formulation is a minimax decision-theoretic perspective that treats the choice of measure as an adversary's move.

A more tractable variant restricts to a subfamily $\mathcal{P} \subseteq \{P_{Y_1, Y_2}\}$:

$$\max_{P \in \mathcal{P}} \min_{f_P} \mathbb{E}_P[(f_P(Y_1) - Y_2)^2].$$

Natural subfamilies (in increasing order of restriction):

- All joint distributions of (Y_1, Y_2) .
- Product measures: Y_1 and Y_2 independent.
- I.i.d. subfamily: $P = P_Y \otimes P_Y$.
- Parametric time-series models (e.g. AR(1)).

On the i.i.d. subfamily the irreducible floor is $\text{Var}Y$, achieved by $f_P^*(Y_1) = \mathbb{E}Y$ — but this minimum depends on P through $\mathbb{E}Y$, so the maxi-min over i.i.d. measures is $\sup_{P_Y} \text{Var}_{P_Y} Y$, which is unbounded without moment constraints.

Restrict to the i.i.d. subfamily: $P = P_Y \otimes P_Y$, so Y_1 and Y_2 are independent with the same marginal distribution. Under i.i.d., $\mathbb{E}_P[Y_2 \mid Y_1] = \mathbb{E}_P Y_2 = \mathbb{E}Y$ (a constant).

Predictor 1: $f(Y_1) = Y_1$ (**identity, a statistic**).

$$\begin{aligned} \mathbb{E}[(Y_1 - Y_2)^2] &= \mathbb{E}[Y_1^2 - 2Y_1Y_2 + Y_2^2] \\ &= \mathbb{E}Y_1^2 - 2(\mathbb{E}Y)^2 + \mathbb{E}Y_2^2 \quad (\text{independence}) \\ &= 2\mathbb{E}Y^2 - 2(\mathbb{E}Y)^2 = 2\text{Var}Y. \end{aligned}$$

The identity function is a statistic: it does not depend on the measure. $\text{MSE} = 2\text{Var}Y$.

Predictor 2: $f_P(Y_1) = \mathbb{E}_P[Y_2 \mid Y_1] = \mathbb{E}Y$ (**measure-dependent constant**).

$$\mathbb{E}[(\mathbb{E}Y - Y_2)^2] = \text{Var}Y.$$

This is the measure-specific optimal predictor. $\text{MSE} = \text{Var}Y$, exactly half of the identity predictor's MSE.

What this illustrates. The optimal predictor f_P^* is twice as good as the statistic Y_1 on the i.i.d. subfamily. But f_P^* requires knowing $\mathbb{E}Y$, which depends on P . In practice, $\mathbb{E}Y$ is unknown and must be estimated. The plug-in predictor — replace $\mathbb{E}Y$ by $\bar{Y}$ or Y_1 — recovers the statistic and pays the price: MSE doubles.

Remark 17. The conditional expectation $\mathbb{E}[Y_2 \mid Y_1]$ does not characterise the full joint distribution of (Y_1, Y_2) : it is one operator of the joint distribution, and many distributions share the same conditional expectation (recall: conditional expectation is a bivariate operator). For the purpose of optimal point prediction, knowing $\mathbb{E}[Y_2 \mid Y_1]$ is sufficient — we do not need the full measure.

3.6 Connection to regression

The same structure appears in regression. The conditional variance $\text{Var}(\hat{\beta}_1 \mid \mathbf{X}) = \sigma^2/S_{XX}$ is a random variable (under homoscedasticity) with a single unknown ingredient: the parameter σ^2 . The function $g = \sigma^2$ is known only up to the parameter — it is the analogue of $\mathbb{E}Y$ in the prediction example.

To predict $\text{Var}(\hat{\beta}_1 \mid \mathbf{X})$, we estimate σ^2 by $\hat{\sigma}^2$ (plug-in) and substitute. This introduces the same gap as replacing $\mathbb{E}Y$ by Y_1 : the predictor is now a statistic, but it pays an additional plug-in cost beyond the irreducible floor.